

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS 8 / 8 completed

Character

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes

PROGRAM EDITOR

Java Run Save

```
1 public class Main {  
2     public static void main(String[] args) {  
3         Character hero = new Hero("Hero");  
4         Character villain = new Villain("Villain");  
5         Character monster = new Monster("Monster");  
  
6  
7         hero.attack(monster);  
8         villain.attack(hero);  
9         monster.attack(villain);  
10    }  
11 }  
12 interface Character {  
13     void attack(Character target);  
14     void takeDamage(int dmg);  
15     void display();  
16 }  
17 abstract class BaseCharacter implements Character {  
18     String name;  
19     int health;  
20     BaseCharacter(String name, int health) {  
21         this.name = name;
```

OUTPUT

Hero attacks!

INPUT

Your INPUT goes here!

31°C Partly cloudy Search [Taskbar icons: File Explorer, Edge, Word, PowerPoint, Outlook, Teams, OneDrive, etc.] ENG IN 44% 08:25 PM 12-08-2026

Windows taskbar showing the search bar, task view button, and various application icons including Edge, Word, and Excel. The system tray on the right displays the date and time as 08:25 PM on 12-08-2026, along with network, volume, and battery status indicators.

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Ask Gemini

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QUESTIONS 8 / 8 completed

TreeMap

TREEMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Map;
2 import java.util.TreeMap;
3 public class Main{
4     public static void main(String[] args){
5         TreeMap<String,Integer>map=new TreeMap<>();
6         map.put("Apple",10);
7         map.put("Banana",25);
8         map.put("Date",30);
9         map.put("Cherry",15);
10        for(Map.Entry<String,Integer>entry:map.entrySet()){
11            System.out.println(entry.getKey()+" "+entry.getValue());
12        }
13    }
14 }
```

OUTPUT

Apple:10

INPUT

Your INPUT goes here!

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QUESTIONS 8 / 8 completed

PriorityQueue

PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.PriorityQueue;
2 public class Main{
3     public static void main(String[] args){
4         PriorityQueue<Integer>pq=new PriorityQueue<>();
5         pq.add(10);
6         pq.add(20);
7         pq.add(15);
8         pq.add(5);
9         System.out.println("PriorityQueue:"+pq);
10    }
11 }
```

OUTPUT

PriorityQueue:[5, 10, 15, 20]

INPUT

Your INPUT goes here!

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